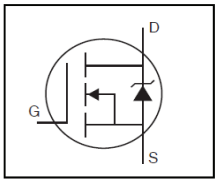
- Surface Mount
- Advanced Process Technology
- Ultra Low On-Resistance
- Dynamic dv/dt Rating
- Fast Switching
- Fully Avalanche Rated
- Lead-Free

HEXFET® Power MOSFET



V_{DSS}	55V
$R_{DS(on)}$	0.065Ω
I_D	3.1A



SOT-223

G	D	S
Gate	Drain	Source

Description

Fifth Generation HEXFETs utilize advanced processing techniques to achieve extremely low on-resistance per silicon area. This benefit, combined with the fast switching speed and ruggedized device design that HEXFET Power MOSFETs are well known for, provides the designer with an extremely efficient and reliable device for use in a wide variety of applications.

The SOT-223 package is designed for surface-mount using vapor phase, infra red, or wave soldering techniques. Its unique package design allows for easy automatic pick-and-place as with other SOT or SOIC packages but has the added advantage of improved thermal performance due to an enlarged tab for heat sinking. Power dissipation of 1.0W is possible in a typical surface mount application.

Base Part Number	Package Type	Standard Pack		Orderable Part Number
		Form	Quantity	
IRLL024NTRPbF	SOT-223	Tape and Reel	2500	IRLL024NTRPbF

Absolute Maximum Ratings

Symbol	Parameter	Max.	Units
I_D @ T_A = 25°C	Continuous Drain Current, V_{GS} @ 10V **	4.4	A
I_D @ T_A = 25°C	Continuous Drain Current, V_{GS} @ 10V *	3.1	
I_D @ T_A = 70°C	Continuous Drain Current, V_{GS} @ 10V *	2.5	
I_{DM}	Pulsed Drain Current ①	12	
P_D @T_A = 25°C	Maximum Power Dissipation (PCB Mount) **	2.1	W
P_D @T_A = 25°C	Maximum Power Dissipation (PCB Mount) *	1.0	
	Linear Derating Factor (PCB Mount) *	8.3	mW/°C
V_{GS}	Gate-to-Source Voltage	± 16	V
E_{AS}	Single Pulse Avalanche Energy (Thermally Limited) ②	120	mJ
I_{AR}	Avalanche Current ①	3.1	A
E_{AR}	Repetitive Avalanche Energy ①*	0.1	mJ
dv/dt	Peak Diode Recovery dv/dt ③	5.0	V/ns
T_J T_{STG}	Operating Junction and Storage Temperature Range	-55 to + 150	°C

Thermal Resistance

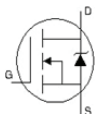
Symbol	Parameter	Typ.	Max.	Units
$R_{\theta JA}$	Junction-to-Ambient (PCB Mount, steady state) *	90	120	°C/W
$R_{\theta JA}$	Junction-to-Ambient (PCB Mount, steady state) **	50	60	

* When mounted on FR-4 board using minimum recommended footprint.
** When mounted on 1 inch square copper board, for comparison with other SMD devices.

Electrical Characteristics @ T_J = 25°C (unless otherwise specified)

	Parameter	Min.	Typ.	Max.	Units	Conditions
$V_{(BR)DSS}$	Drain-to-Source Breakdown Voltage	55	—	—	V	V_{GS} = 0V, I_D = 250µA
$\Delta V_{(BR)DSS}/\Delta T_J$	Breakdown Voltage Temp. Coefficient	—	0.048	—	V/°C	Reference to 25°C, I_D = 1mA
$R_{DS(on)}$	Static Drain-to-Source On-Resistance	—	—	0.065	Ω	V_{GS} = 10V, I_D = 3.1A ④
		—	—	0.080		V_{GS} = 5.0V, I_D = 2.5A ④
		—	—	0.100		V_{GS} = 4.0V, I_D = 1.6A ④
$V_{GS(th)}$	Gate Threshold Voltage	1.0	—	2.0	V	V_{DS} = V_{GS}, I_D = 250µA
gfs	Forward Trans conductance	3.3	—	—	S	V_{DS} = 25V, I_D = 1.9A
I_{DSS}	Drain-to-Source Leakage Current	—	—	25	µA	V_{DS} = 55 V, V_{GS} = 0V
		—	—	250		V_{DS} = 44V,V_{GS} = 0V,T_J =125°C
I_{GSS}	Gate-to-Source Forward Leakage	—	—	100	nA	V_{GS} = 16V
	Gate-to-Source Reverse Leakage	—	—	-100		V_{GS} = -16V
Q_g	Total Gate Charge	—	10.4	15.6	nC	I_D = 1.9A
Q_{gs}	Gate-to-Source Charge	—	1.5	2.3		V_{DS} = 44V
Q_{gd}	Gate-to-Drain Charge	—	5.5	8.3		V_{GS} = 5.0V , See Fig. 6 and 13 ④
$t_{d(on)}$	Turn-On Delay Time	—	7.4	—	ns	V_{DD} = 28V
t_r	Rise Time	—	21	—		I_D = 1.9A
$t_{d(off)}$	Turn-Off Delay Time	—	18	—		R_G= 24Ω
t_f	Fall Time	—	25	—		R_D= 15Ω, See Fig. 10 ④
C_{iss}	Input Capacitance	—	510	—	pF	V_{GS} = 0V
C_{oss}	Output Capacitance	—	140	—		V_{DS} = 25V
C_{rss}	Reverse Transfer Capacitance	—	58	—		f = 1.0MHz, See Fig. 5

Source-Drain Ratings and Characteristics

	Parameter	Min.	Typ.	Max.	Units	Conditions
I_S	Continuous Source Current (Body Diode)	—	—	3.1	A	MOSFET symbol showing the integral reverse p-n junction diode. 
I_{SM}	Pulsed Source Current (Body Diode) ①	—	—	12		
V_{SD}	Diode Forward Voltage	—	—	1.0	V	T_J = 25°C,I_S = 1.9A,V_{GS} = 0V ④
t_{rr}	Reverse Recovery Time	—	39	58	ns	T_J = 25°C ,I_F = 1.9A
Q_{rr}	Reverse Recovery Charge	—	63	94	nC	di/dt = 100A/µs ④

Notes:

① Repetitive rating; pulse width limited by max. junction temperature. (See fig. 11)

② starting T_J = 25°C, L = 25mH, R_G = 25Ω, I_{AS} = 3.1A (See fig. 12)

③ I_{SD} ≤ 1.9A, di/dt ≤ 270A/µs, V_{DD} ≤ $V_{(BR)DSS}$, T_J ≤ 150°C.

④ Pulse width ≤ 300µs; duty cycle ≤ 2%.

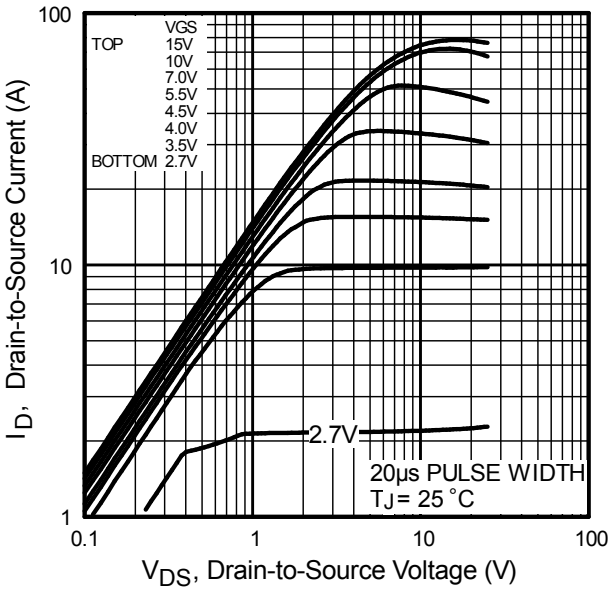


Fig. 1 Typical Output Characteristics

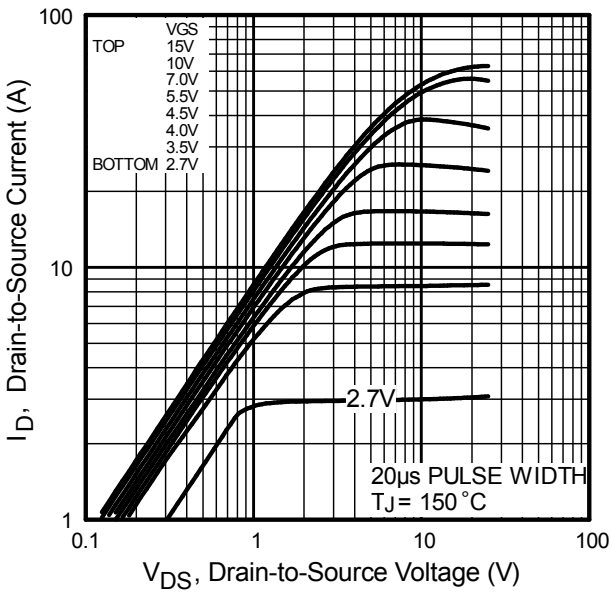


Fig. 2 Typical Output Characteristics

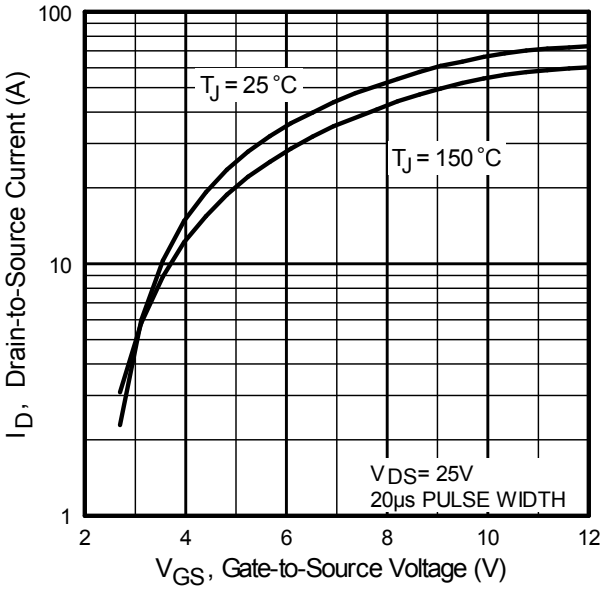


Fig. 3 Typical Transfer Characteristics

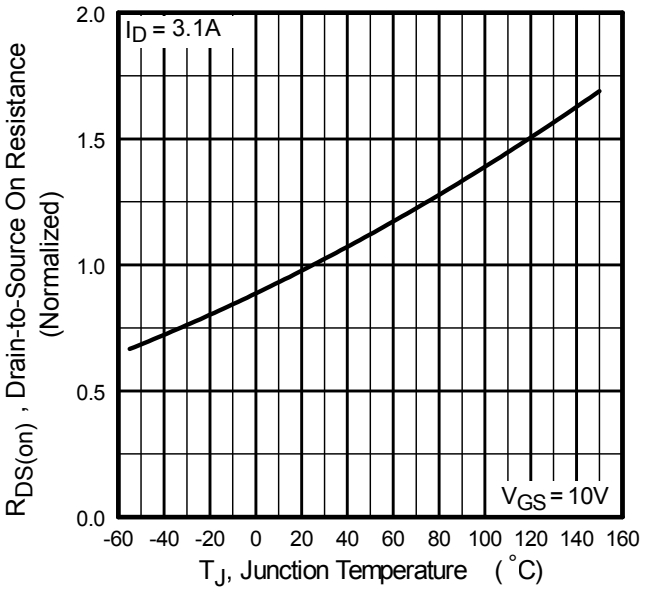


Fig. 4 Normalized On-Resistance vs. Temperature

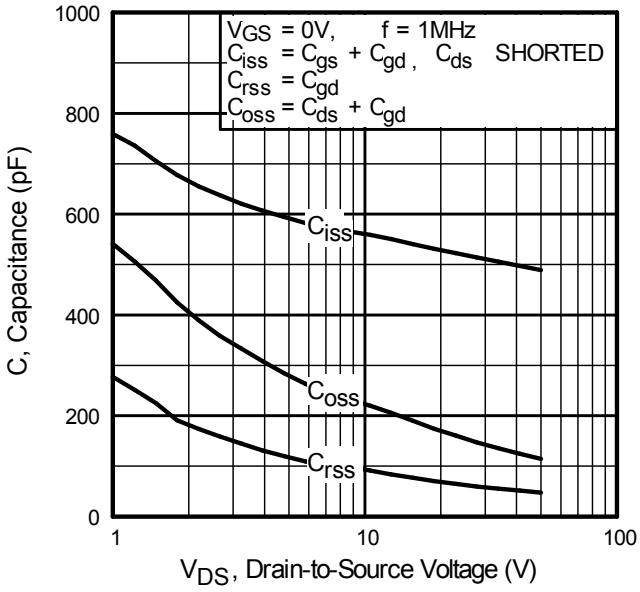


Fig 5. Typical Capacitance vs. Drain-to-Source Voltage

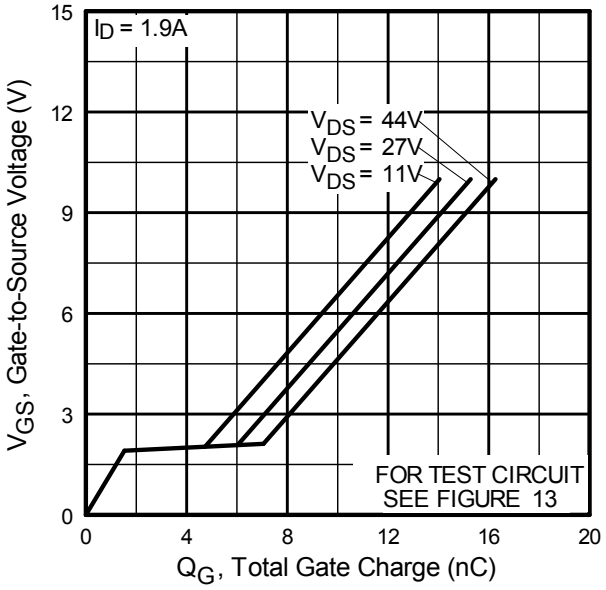


Fig 6. Typical Gate Charge vs. Gate-to-Source Voltage

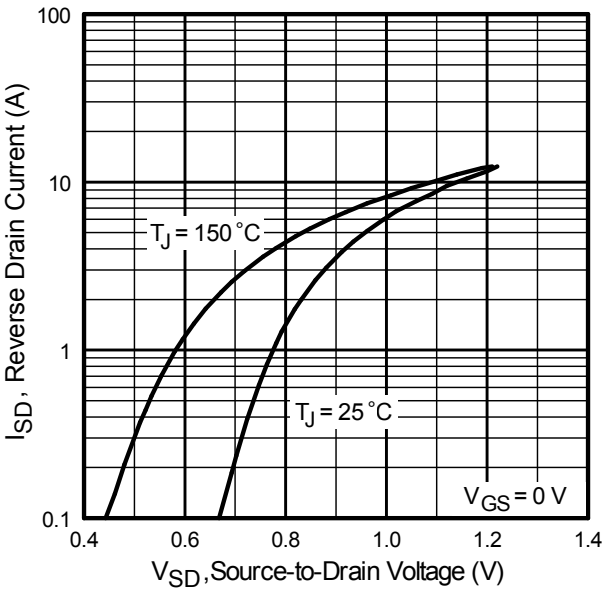


Fig. 7 Typical Source-to-Drain Diode Forward Voltage

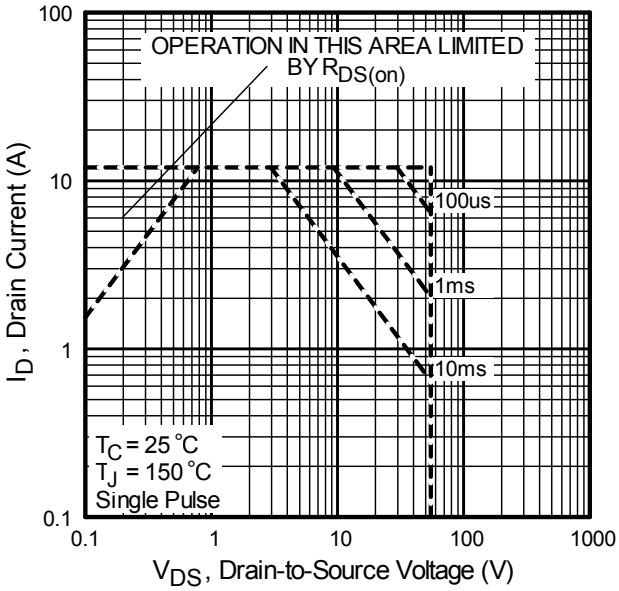


Fig 8. Maximum Safe Operating Area

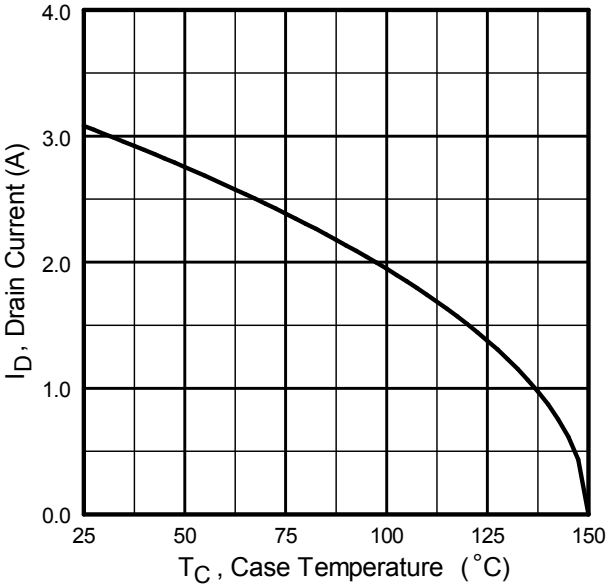


Fig 9. Maximum Drain Current Vs. Case Temperature

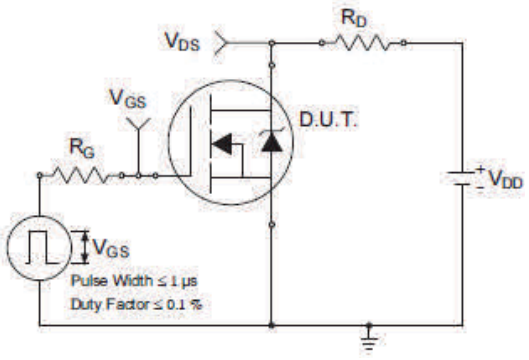


Fig 10a. Switching Time Test Circuit

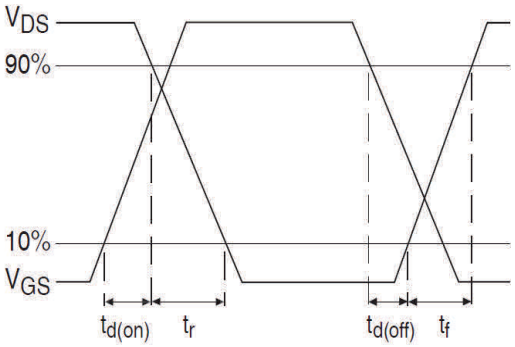


Fig 10b. Switching Time Waveforms

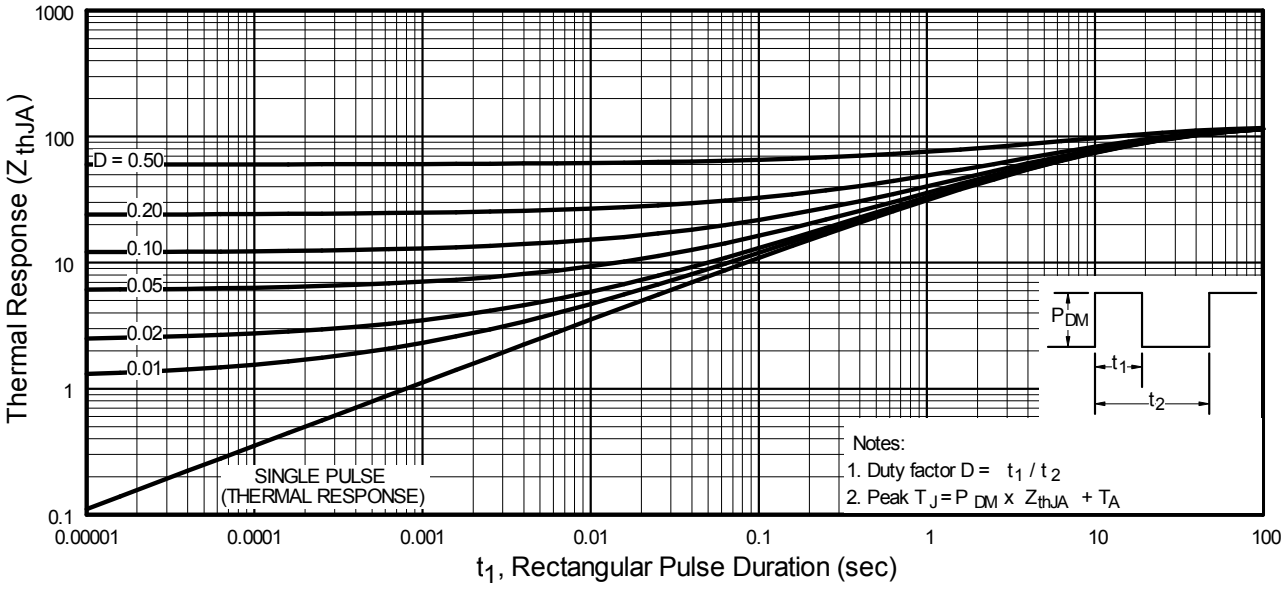


Fig 11. Maximum Effective Transient Thermal Impedance, Junction-to-Ambient

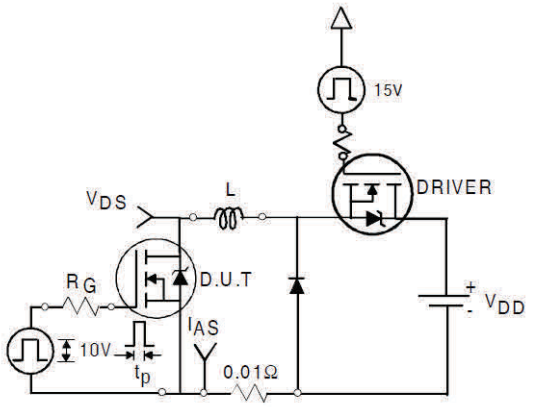


Fig 12a. Unclamped Inductive Test Circuit

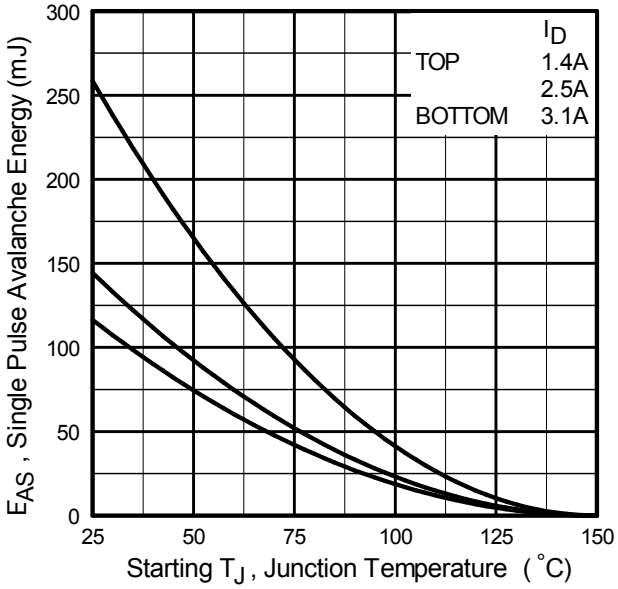


Fig 12c. Maximum Avalanche Energy vs. Drain Current

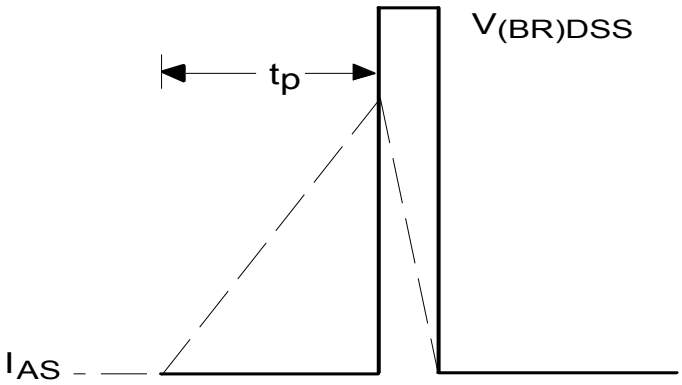


Fig 12b. Unclamped Inductive Waveforms

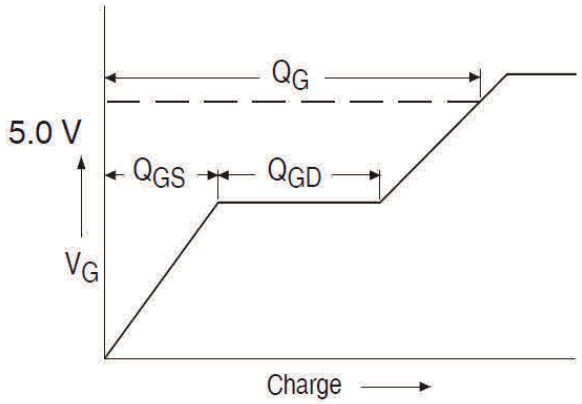


Fig 13a. Basic Gate Charge Waveform

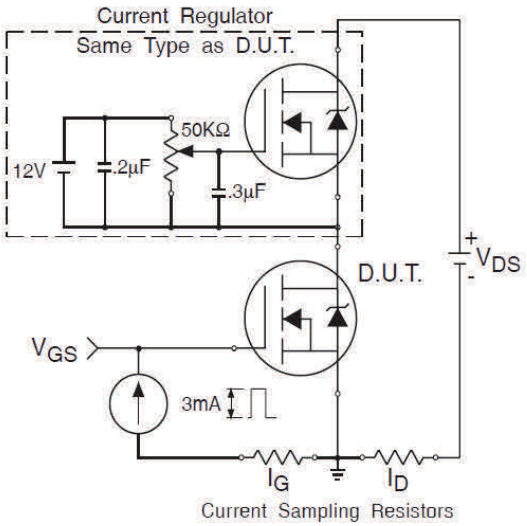
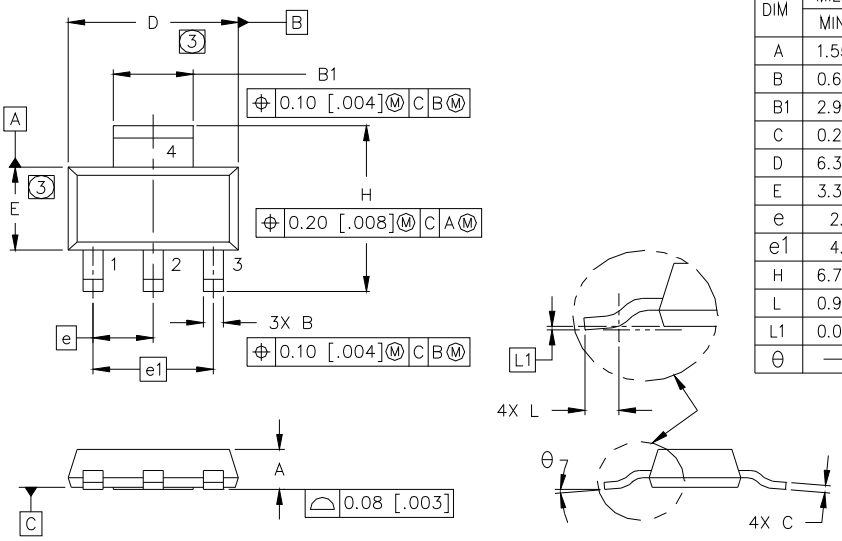
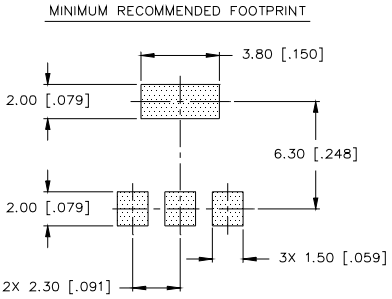


Fig 13b. Gate Charge Test Circuit

SOT-223 (TO-261AA) Package Outline (Dimensions are shown in millimeters (inches)



DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	1.55	1.80	.061	.071
B	0.65	0.85	.026	.033
B1	2.95	3.15	.116	.124
C	0.25	0.35	.010	.014
D	6.30	6.70	.248	.264
E	3.30	3.70	.130	.146
e	2.30 BSC		.0905 BSC	
e1	4.60 BSC		.181 BSC	
H	6.71	7.29	.264	.287
L	0.91	—	.036	—
L1	0.061 BSC		.0024 BSC	
θ	—	10°	—	10°



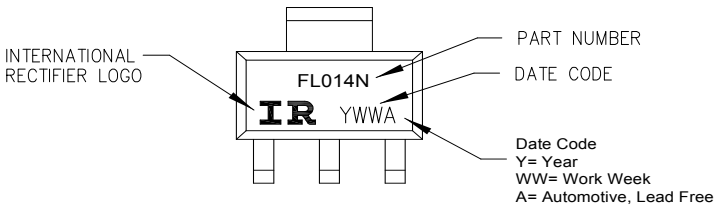
LEAD ASSIGNMENTS

1 = GATE
2 = DRAIN
3 = SOURCE
4 = DRAIN

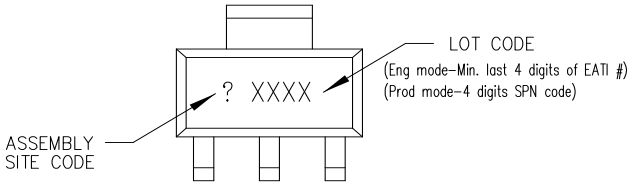
NOTES:

1. DIMENSIONING & TOLERANCING PER ASME Y14.5M–1994.
2. CONTROLLING DIMENSION: INCH.
3. DIMENSIONS DO NOT INCLUDE MOLD FLASH.
4. OUTLINE CONFORMS TO JEDEC OUTLINE TO–261AA.
5. DIMENSIONS ARE SHOWN IN MILLIMETERS [INCHES].

SOT-223(TO-261AA) Part Marking Information

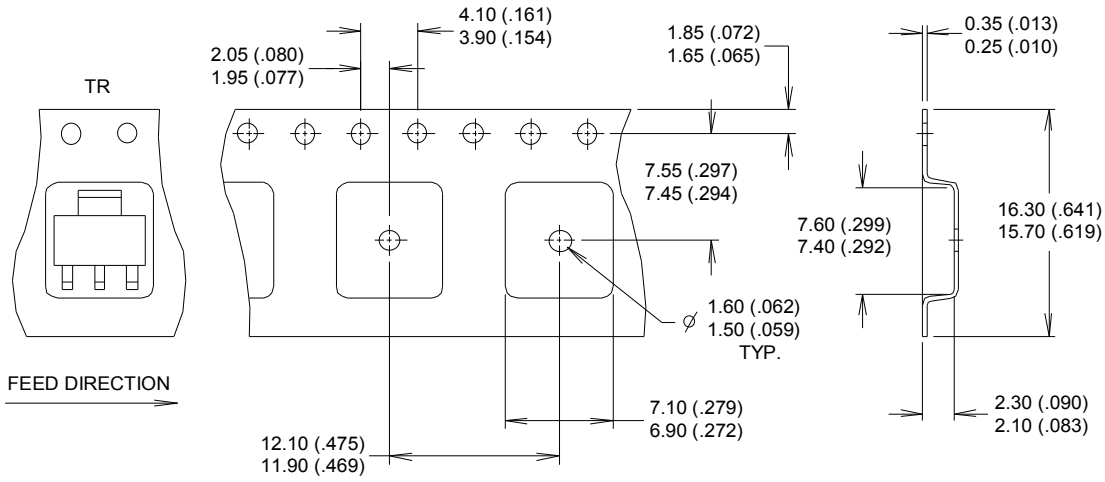


TOP MARKING

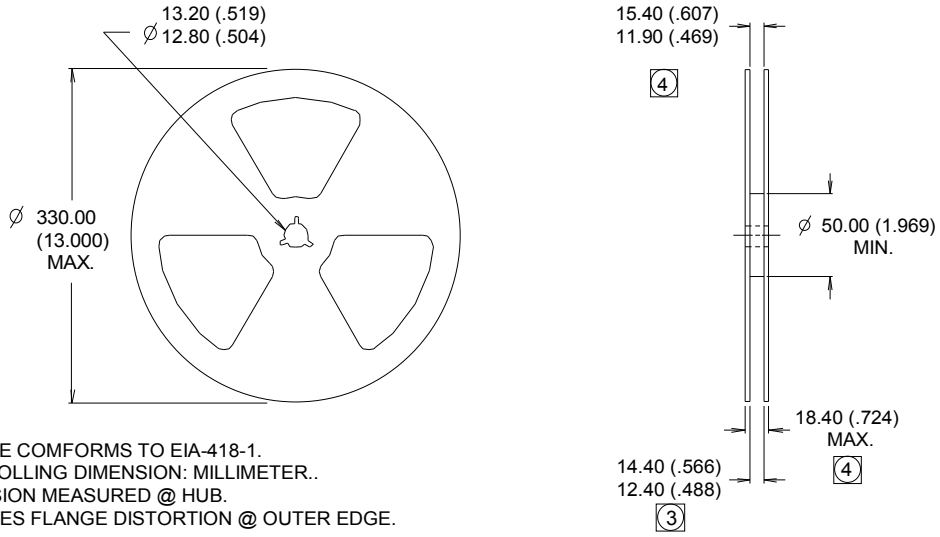


BOTTOM MARKING

SOT-223(TO-261AA) Tape and Reel (Dimensions are shown in millimeters (inches)



NOTES :
1. CONTROLLING DIMENSION: MILLIMETER.
2. OUTLINE CONFORMS TO EIA-481 & EIA-541.
3. EACH Ø330.00 (13.00) REEL CONTAINS 2,500 DEVICES.



NOTES :
1. OUTLINE COMFORMS TO EIA-418-1.
2. CONTROLLING DIMENSION: MILLIMETER..
3. DIMENSION MEASURED @ HUB.
4. INCLUDES FLANGE DISTORTION @ OUTER EDGE.

Note: For the most current drawing please refer to Infineon's web site www.infineon.com

Qualification Information

Qualification Level	Industrial (per JEDEC JESD47F) †	
Moisture Sensitivity Level	SOT-223	MSL1 (per JEDEC J-STD-020D) †
RoHS Compliant	Yes	

† Applicable version of JEDEC standard at the time of product release (04/27/2004).

Revision History

Date	Comments
01/28/2019	• Updated datasheet with corporate template. • Added disclaimer on last page. • Corrected part number from"IRLL024NPbF" to "IRLL024NTRPbF"-all pages

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